

Participant Payout Analysis

Supplier–Retailer Negotiation Experiment

July 15, 2026

Payment rule

Each participant receives a baseline of €5 plus a bonus of 5% of their realized profit (if positive), computed after the demand draw:

$$\pi_S = w \cdot \min(q, D) - PC \cdot q, \quad \pi_R = (RP_{eff} - w) \cdot \min(q, D),$$

with $PC = 1$, $RP \in \{4, 5\}$, $w \in [1, 5]$, $q \leq 100$, and $RP_{eff} = RP + 0.02$ per passed slider (capped at +1). Demand D is the *average of ten* draws from $U[0, 100]$, so $D \approx 50 \pm 9$; extreme demand is effectively impossible.

Bounds

	Min	Theoretical max	Realistic max
Supplier	€5	€25	$\approx$ €14
Retailer	€5	€30	$\approx$ €18
Per pair	€10	€35	$\approx$ €28

The minimum is always the baseline (negative profit earns no bonus). The theoretical maxima require $D = 100$, i.e. all ten draws equal to 100 (probability $\sim 10^{-20}$); the ten-draw average therefore insures the budget against extreme payouts. In a 200,000-participant simulation the largest observed payment was €18.09 (95th percentile: €12.16).

Expected payouts (90% agreement rate)

Monte Carlo with deals at or dispersed around the Nash terms (RP = 4: 2.80€/75; RP = 5: 3.33€/80), 10% of participants at baseline only:

Sliders passed (quality checks)	Supplier	Retailer
None	€8.79	€8.60
Half	€8.98	€9.82
All 50	€8.98	€11.05

Central estimate: $\approx$ €8.90 per participant (median €9.10), i.e. $\approx$ €17.85 per two-human pair. Budget rule of thumb: €9 per head, p95 $\approx$ €12, practical ceiling $\approx$ €18.

Sensitivities

- **Slider effort** is the main lever: each passed slider adds $\approx \text{€}0.05$ to the Retailer's payment (full effort: $+\text{€}2.5$), while paying the Supplier nothing.
- **Supplier payouts are flat** ($\approx \text{€}8.8\text{--}9.0$) across bargaining outcomes near the Nash terms.
- **AI-retailer sessions** pay only the human Supplier: $\approx \text{€}8.60$ per participant all-in.